

# Answers

1. Question: Name two key relationships between the physical quantities that are important in a study of the resistance of ships. Why is each important?

The answer should include the following:

- Froude number. Important for wave-making resistance, particularly at higher speeds.
- Reynolds number. Important for frictional resistance. Important at lower speeds.
- Comparisons of model and full scale results made at the same value of Froude number; concept of corresponding Froude number.

2. Question: Discuss the various types of resistance encountered by a ship. How can they be reduced? What is their relative importance at different speed regimes?

The answer should include the following:

- Wave-making resistance. High speed. Use of fine forms forward and bulbous bows.
- Frictional resistance. Relatively more important at lower speeds. Reduce wetted surface area. Keep the underwater hull clean and free of fouling.
- Eddy making resistance. Avoid sudden changes in direction of the fluid flow.
- Appendage resistance. Avoid protrusions from the ship. For essential appendages, use streamlined shapes (e.g. rudders) and align with the general direction of flow (e.g. shaft brackets).
- Wind resistance. Depends upon the relative speed of the ship to the air. In a head wind, resistance will become large. Streamline the above water hull and the superstructure.

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3. Question: What do you understand by roughness? How does it come about? How can it be reduced?

The answer should include the following:

- Concepts of laminar and turbulent flow.
- The boundary layer, increasing in thickness from the bow to the stern.
- Arises from irregularities on the wetted surface plate overlaps, welding, pitting and plate waviness, marine growth.
- To reduce:
  - a. Ensure surface is as smooth as possible, as built.
  - b. Preserve plating as it is fabricated to reduce initial corrosion.
  - c. Use good anti-corrosion coatings to reduce corrosion in service.
  - d. Possibly install a cathodic protection system.
  - e. Reduce fouling by good outer bottom coatings and regular dockings.